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M 405 CHEMICAL INJECTION SYSTEM

1. SCOPE OF WORK

The Bidder shall supply the following in accordance with part 4, Subsection 1.2. “Summary of Work”.

The Chemical Injection system shall be provided for each Block (Block 2, 3 & 4)

The equipment for chemical injection for each block shall include but not be limited to the following for each condensate system and HRSG.

Carbohydrazide feed equipment consisting of a two (2) chemical solution tank (diluted carbohydrazide tank and concentrated carbohydrazide tank), 1 x100 % transferring pump from concentrated carbohydrazide tank to diluted carbohydrazide tank, 2 x 100 % pumps to condensate dosing and deaerator dosing completed with diaphragm metering, agitator, controls, piping and accessories up to the injection point in the condensate system.

Ammonia feed consisting of a chemical solution tank, 2 x 100 % pumps to condensate dosing and deaerator dosing completed with diaphragm metering, agitator, controls, piping and accessories up to the injection point in the condensate system.

Phosphate feed equipment consisting of chemical solution tank, 2 x 100 % pumps completed with diaphragm metering, agitator, dissolving basket, controls, piping and accessories up to the injection point to each HRSG unit.

Complete Instrumentation and Controls and Electrical Systems including devices, control panels, wiring, piping, tubing and valves as detailed in Clause 7 and Clause 8 herein.

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”.

The following Codes and Standards extracted from Subsection 2.5 are specifically applicable to the design manufacture and testing of the equipment in this specification.

ANSI	American National Standards Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society for testing and Material
HIS	Hydraulic Institute Standards
IEG	International Electro technical Commission
IEEE	Institute of Electrical and Electronics Engineers
NEMA	National Electrical Manufactures Association

3. DESIGN AND OPERATING CONDITIONS

The system shall be designed to operate automatically. Injection of carbonylhydrazide shall be controlled from the condensate flow and the injection of ammonia from the condensate pH. The necessary instruments and controls shall be provided to give a 4 - 20 mA DC signal to control the pump stroke. Under normal circumstances, the operator will set the stroke of the pumps as necessary. A portable hand pump shall be supplied for carbonylhydrazide and ammonia solution transfer ~~from the chemical storage tank to the solution tank~~. Phosphate is a solid would be placed in the dissolving basket in the phosphate chemical tank.

3.1 Design Conditions

The following design conditions describe the requirement for each block only.

3.1.1 Dilute Carbonylhydrazide

Number of Solution Tanks	1
Minimum solution Tank Capacity	Bidder determine
Solution Tank Design Pressure	Atmospheric
Number of Solution Pumps	2 (1 as 100% spare)

Material of Construction for :

Solution Tank, Valves and Piping Stainless steel

Number of Agitators 1

3.1.2 Ammonia Feed Equipment

Number of Solution Tanks 1

Minimum solution Tank Capacity Bidder determine

Solution Tank Design Pressure Atmospheric

Number of Solution Pumps 2 (1 as 100% spare)

Material of Construction for :

Solution Tank, Valves and Piping Stainless steel

Number of Agitators 1

3.1.3 Phosphate Feed Equipment

Number of Solution Tanks 1

Minimum solution Tank Capacity Bidder determine

Solution Tank Design Pressure Atmospheric

Number of Solution Pumps 2 (1 as 100% spare) for each pressure

Material of Construction for :

Solution Tank, Valves and Piping Stainless steel

Number of Agitators 1

3.1.4 Concentrate Carbohydrazide

Number of Solution Tanks 1

Minimum solution Tank Capacity Bidder determine

Solution Tank Design Pressure Atmospheric

Number of Solution Pumps 1

Material of Construction for :

Solution Tank, Valves and Piping Stainless steel

Number of Agitators 1

3.2 Equipment Sizing

Sizing of tanks and pumps is based on the following The Bidder shall satisfy himself that all equipment sizing is correct.

3.2.1 Carbohydrazide Feed

Feed Rate	2 ppm x O2 in the condensate.
Solution Concentration	1 % - 2%
Tank Storage Capacity	72 h

3.2.2 Ammonia Feed

Feed Rate	3.3 ppm
Solution Concentration	10%
Tank Storage Capacity	48 h

3.2.3 Phosphate Feed

Feed Rate	Slug close as required.
Solution Concentration	1%
HRSG Blowdown Rate	Bidder determine
Tank Storage Capacity	72 h

3.2.4 Concentrate Carbohydrazide Feed

Feed Rate	Bidder determine.
Solution Concentration	Bidder determine
Tank Storage Capacity	6 times

4. DETAILED REQUIREMENTS

4.1 Pumps

Metering pumps shall be furnished in accordance with Part 4, Specification M-009 (all metering pumps).

The pump, motor and speed reducer for each pump unit shall be mounted on a skid common with its respective tank. Each pump shall be completely piped. Pumps shall be located in front of (not under) their respective mixing tanks. Pumps, motors, piping, tanks, and accessories shall be skid mounted.

While the pump are in operation from 0 to 100% capacity, each pump shall be capable of delivering full capacity against maximum pressure at the injecting point of chemical. The pump flow accuracy shall be within $\pm 3\%$ of the total rated below.

Pump material shall be of the best suitable material and inner parts contacting the chemicals shall be of stainless steel.

4.2 Description of Tanks

The tanks shall be used as suction tanks from which the chemical feed pumps will receive their supply.

Each tank shall be of the vertical cylindrical type and shall be furnished complete with welded cover with hinged door, sight gauge, valved water fill and drain connections etc. The disodium phosphate tank shall be provided with stainless steel dissolving basket.

Thickness of each tank shall be as required for the service conditions.

Each tank shall be furnished with a glass gauge, attached on the outside of the tank, calibrated in liters, high-level and low-level alarm switches, and low-level pump shutoff switches with supports and other necessary accessories.

Shop fabricated tanks shall be supplied and installed in accordance with Part 4, Specification M-012.

4.3 Mixer

The mixer shall be of propeller type direct motor driven.

The mixer shall be capable of agitating the maximum capacity of feed solutions in the tank.

The shaft and propeller shall be stainless steel.

4.4 Piping and Valves

Pump relief valve shall be set at the maximum pressure of chemical injection point, and the pump and accessories shall be designed based on the above pressure condition.

The Bidder shall provide the interconnecting piping between tanks and pump, and shall provide the chemical feed piping between pump and chemical injection points, including drain and vent piping with stop and check valves, relief valves, pressure gauges, pipe supports and all necessary pertinents.

The piping shall be in conformity with Part 4, Technical Specification M-001.

The Valves shall be in accordance with Part 4, Technical Specifications M-004 and M-005.

4.5 Miscellaneous

One (1) ASTM 304 stainless steel dissolving basket shall be supplied to be used for disodium phosphate tank.

Motor-driven agitators with ASTM 304 stainless steel shaft and propellers shall be mounted on each chemical solution tank.

All hand pumps piping and valves shall be furnished to feed the chemicals to the appropriate tanks.

Measuring cylinder shall be included on ammonium solution and carbohydrazide tanks Each cylinder shall be mounted on its respective tank with inlet and outlet valves, and all accessories.

The ammonia measuring cylinder shall be approximately of 30% solution and shall be capable of measuring two litres accurately.

The carbohydrazide measuring cylinder shall be approximately 0.5 litre and be capable of measuring 25 cm³ accurately.

All materials not specifically covered herein shall be of the best quality for the service contemplated.

4.6 Painting and Protective Coating

Before shipment, all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter, and shall be shop painted, suitable for the service and environment and touch up or repairing as necessary at site shall be provided. The color of the final coat shall be as per the Bidder's standard. No paints having an asphaltic base shall be used.

The Bidder shall ship a sufficient quantity of paint for touching up where final coat has been damaged during shipment and erection. Paint shall be the same kind and quality as used for each piece of equipment in the shop.

Painting shall be per Part 4, Specification M-001, S-420 and S-430.

5. PERFORMANCE TESTING AND GUARANTEES

The Bidder shall conduct necessary functional test of chemical injection systems during the overall performance test of the plant by checking the cycle water chemistry during the plant performance tests.

5.1 Site Test

Site test will be conducted by the Bidder to prove guaranteed performance.

5.2 Guarantees

Each pump shall be guaranteed to continuously supply the respective chemical at the design flow rates required to maintain the proper water chemistry when the steam turbine is operating based on Gas Turbine peak load. Also refer to Part 4, Technical Specification M-009 Metering Pumps for additional guarantee requirements

6. QUALITY ASSURANCE / QUALITY CONTROL (QA/QC)

Quality Assurance/Quality Control shall conform to the requirements of Part 4, Subsection 2.7.

The quality control procedures as specified in Part 4 Subsection 2.7 shall include but not be limited to the following :

- Packaging and Shipping
- Shop Test Procedures
- Cleaning and Painting

The quality control documentation as specified in Part 4 Subsection 2.7 shall include, but not be limited to the following :

- Shop Test Reports
- All other documentation required by applicable codes and standards

7. INSTRUMENTATION AND CONTROLS

7.1 General

The Bidder shall be responsible for the design, supply and installation of complete instrumentation and controls systems and devices for the Chemical Injection System including, but not be limited to, those listed and described in this Technical Specification.

The Bidder shall be responsible for the design, supply, installation and termination of all interconnecting field cables.

7.2 Local Control System

The Bidder shall furnish and install a local control system for Chemical Injection System.

The local control system shall be basically distributed control system (DCS) unless otherwise specified.

The DCS shall be provided with the operators console and the VDU.

The DCS shall contain manual and automatic control function, graphic monitoring function, alarming function etc., and necessary device to interface to other control system and the Plant and Information and Management System.

All instruments and control devices shall be wired to terminal blocks and piped to terminal points as applicable.

The chemical feed equipment control system shall include, but not be limited to the followings :

Following alarming function shall be provided as a minimum :

- (a) Motor overload
- (b) Loss of control power
- (c) Phosphate tank high/low and low levels
- (d) Ammonium hydroxide tank high/low and low levels
- (e) Carbohydrazide tank high/low and low levels, etc.

Following ON – OFF control shall be provide :

- Ammonium hydroxide pumps
- Carbohydrazide pumps
- Agitator motor

Following modulation control with Manual/Auto station shall be provide with and arrange of 4-20 mA DC control signal:

- Ammonium hydroxide pumps
- Carbohydrazide pumps
- Phosphate pumps.

7.3 Locally Mounted Devices

High and low level alarm switches and low low level pump trip switches on each tank wired to junction boxes.

Level gauges on each tank.

Pressure gauge with chemical seal on the discharge of each pump.

The detailed description of local control panel shall be refereed to Part 4, Technical Specifications of I-001 and I-002.

8. ELECTRICAL SPECIFICATIONS

8.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for the Chemical Injection System. For detailed requirement of electrical equipment, refer to electrical Technical Specifications.

Those equipment shall include, but not be limited to, the followings:

1. Motors.
2. Power panels.
3. Conduits, cables and wiring.
4. All electrical power supply, control, monitoring and instrumentation devices and other appurtenances required for this Specification, shall be provided by the Bidder.

8.2 Motors

For detailed electrical requirements, refer to Part 4, Technical Specification E-210 for low voltage motors and E-211 for medium voltage motors.

8.3 Power Panels

For detailed electrical requirements, refer to Part 4, Technical Specification E-280.

The panel shall have a degree of protection as specified.

8.4 Conduits, Cables and Wiring

All motors and electrical, control, monitoring and instrumentation devices furnished and installed by the Bidder shall be wired by him to the power and local control panel respectively.

The wiring between those panels shall also be included.

The required power, control, instrumentation and thermocouple extension cables shall be provided by the Bidder.

For detailed electrical requirements, refer to Part 4, Technical Specifications E-230 and E-232.

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